---------- Matplotlib ----------

Matplotlib is a popular Python library used for data visualization—it helps you create graphs like line charts, bar charts, histograms, etc.

Matplotlib is a Python library used for creating static, animated, and interactive plots.

Data visualization: It's the graphical representation of data to identify patterns, trends, and insights.

Key Features :- ▪ Multiple plots → line, bar, scatter, histogram ▪ Highly customizable → colors, styles, labels ▪ Works with NumPy & Pandas ▪ Figure & Axes control (OOP style) ▪ Save in formats → PNG, PDF, JPG ▪ Interactive plots (zoom, pan)

1. Line Chart

2. Customize Chart

3. Multiple Plot (Multiple Line & Legends)

4. Bar Chart

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7. Pie Chart

8. Scatter Plot

9. Bubble Plot

10. Box Plot

11. Subplot

12. Matplotlib with Pandas - real data

13. Object-Oriented API

14. 3D Plot

15. Save Charts

In [ ]:

```

```

In [183…

```
# install matplotlib
!pip install matplotlib
```

```
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Requirement already satisfied: cycler>=0.10 in c:\users\satya\anaconda3\lib\site-
packages (from matplotlib) (0.11.0)
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Requirement already satisfied: six>=1.5 in c:\users\satya\anaconda3\lib\site-pack
ages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

In [184…

```python
# import matplotlib
import matplotlib.pyplot as plt

# pyplot is a collection of functions that make matplotlib work as MATLAB
```

In [ ]:

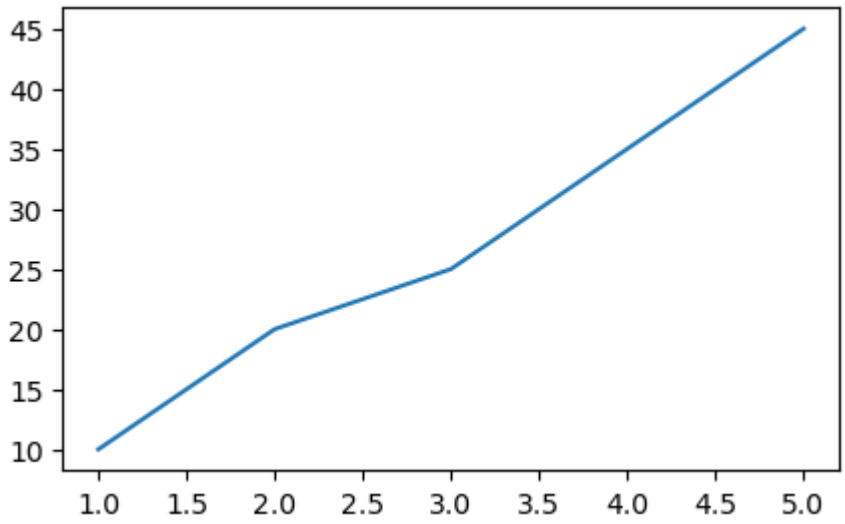
1. Line Chart

A line chart shows how data changes over time by connecting points with a line.

It is mainly used to visualize trends and patterns.

In [118…

```python
plt.figure(figsize=(5,3))  # figure size

# data
x= [1,2,3,4,5]
y= [10,20,25,35,45]

plt.plot(x,y)  # line chart
plt.show()  # display chart in jupyter-notebook
```

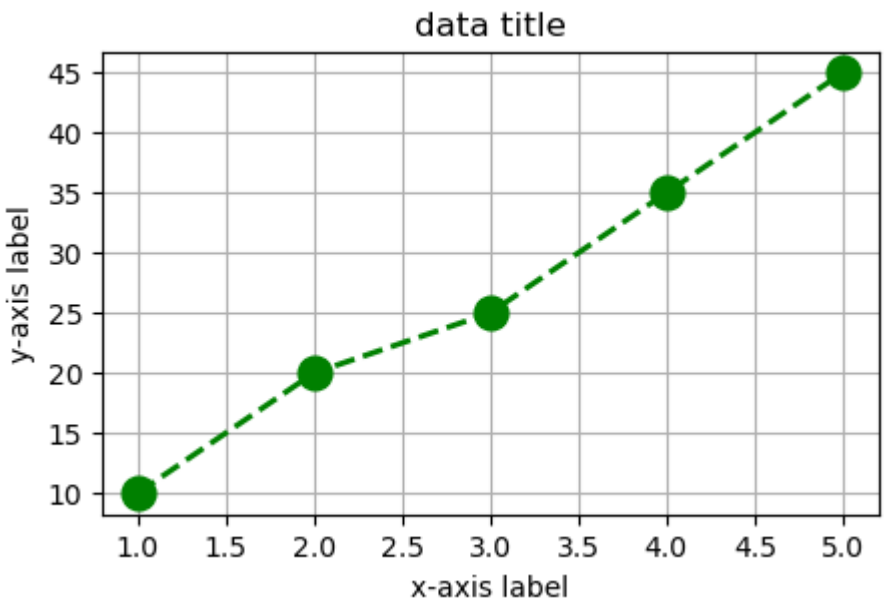


2. Customize Chart

A customized chart is a graph where you modify elements like color, labels, title, size, and style to improve visualization and clarity.

In [119…

```
plt.figure(figsize=(5,3))  # figure size

plt.plot(x,y, color='green', marker='o', linestyle= 'dashed', linewidth=2, marke

plt.title('data title')
plt.xlabel('x-axis label')
plt.ylabel('y-axis label')

plt.grid()  # to show grid on page
plt.show()
```



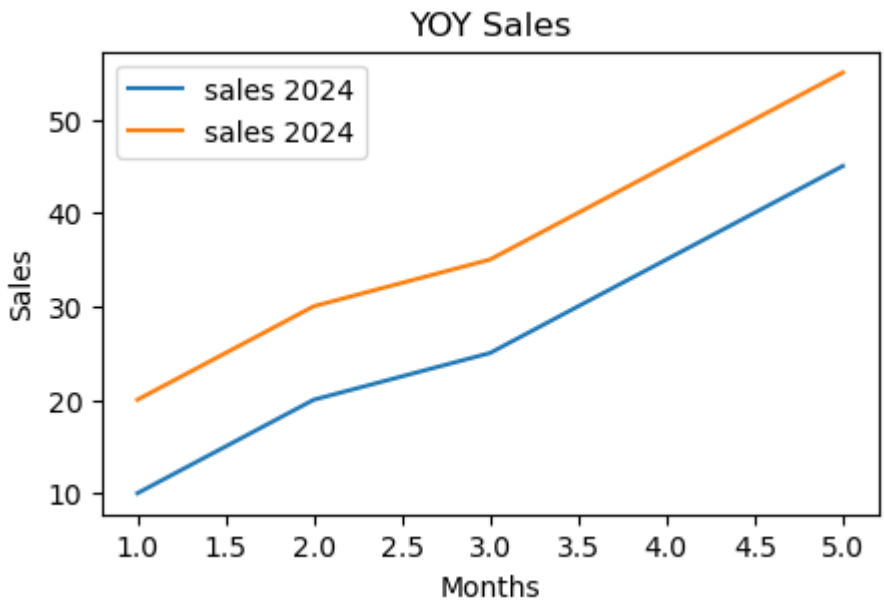
3. Multiple Plot (Multiple Line & Legends)

A multiple line chart shows more than one data line on the same graph to compare trends.

Legends are used to identify each line clearly.

In [120…

```
plt.figure(figsize=(5,3))  # figure size

# data
x= [1,2,3,4,5]
y= [10,20,25,35,45]
z= [20,30,35,45,55]

plt.plot(x,y,label='sales 2024') # ploting y data
plt.plot(x,z,label='sales 2024') # ploting z data

plt.title('YOY Sales')    #YOY- year on year
plt.xlabel('Months')
plt.ylabel('Sales')

plt.legend() # show legends
plt.show()
```

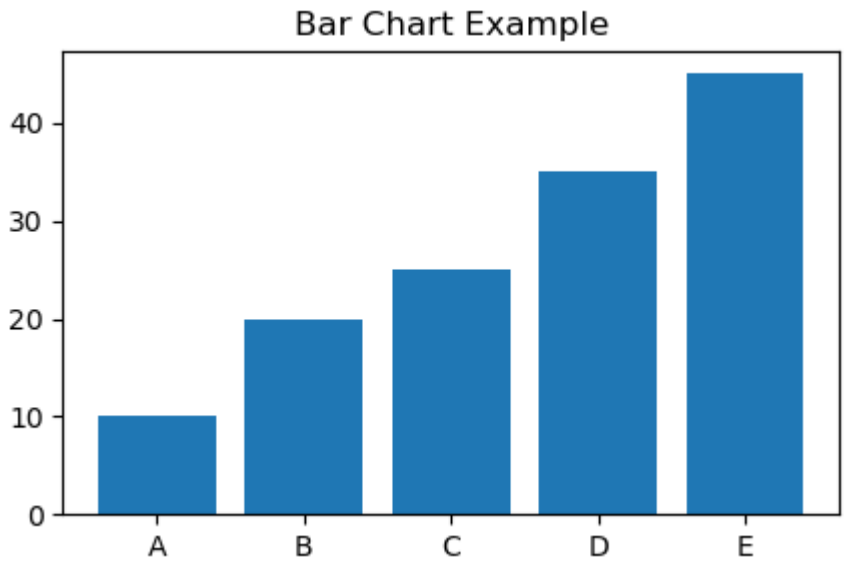


4. Bar Chart

A bar chart is used to compare values across different categories using rectangular bars.

In [38]:

```
plt.figure(figsize=(5,3))  # figure size

# data
x= ['A','B','C','D','E']
y= [10,20,25,35,45]

plt.bar(x,y)
plt.title('Bar Chart Example')
plt.show()
```



5. Countplot

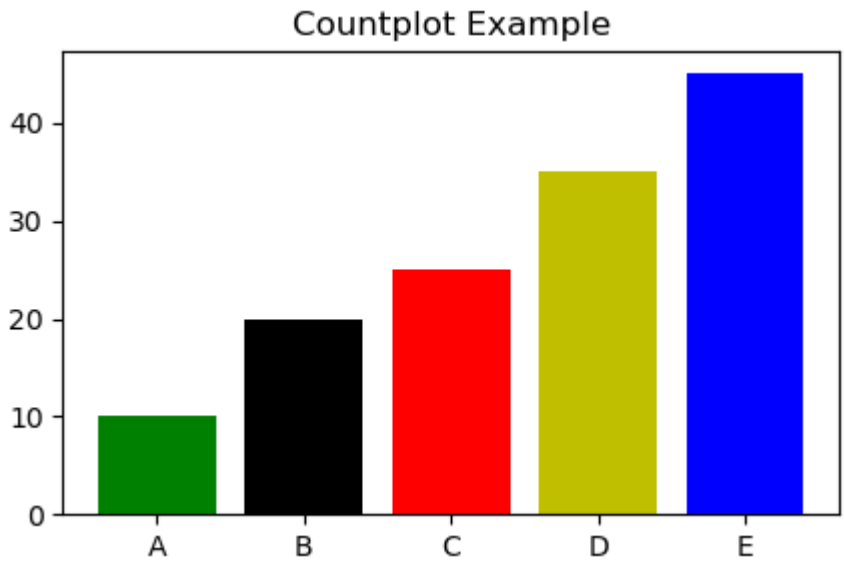
A count plot shows the number of occurrences of each category using bars.

- Shows frequency of categorical data
- No need for manual counting (automatic)
- Created using Seaborn
- Supports **Hue** for sub-category comparison
- Useful in EDA (Exploratory Data Analysis)

Hue means adding a second categorical variable to divide data into sub-groups using different colors.

In [121…

```
plt.figure(figsize=(5,3))  # figure size

# data
x= ['A','B','C','D','E']
y= [10,20,25,35,45]

plt.bar(x,y , color = ['g','k','r','y','b'])  # to show diff-diff colour
plt.title('Countplot Example')
plt.show()
```



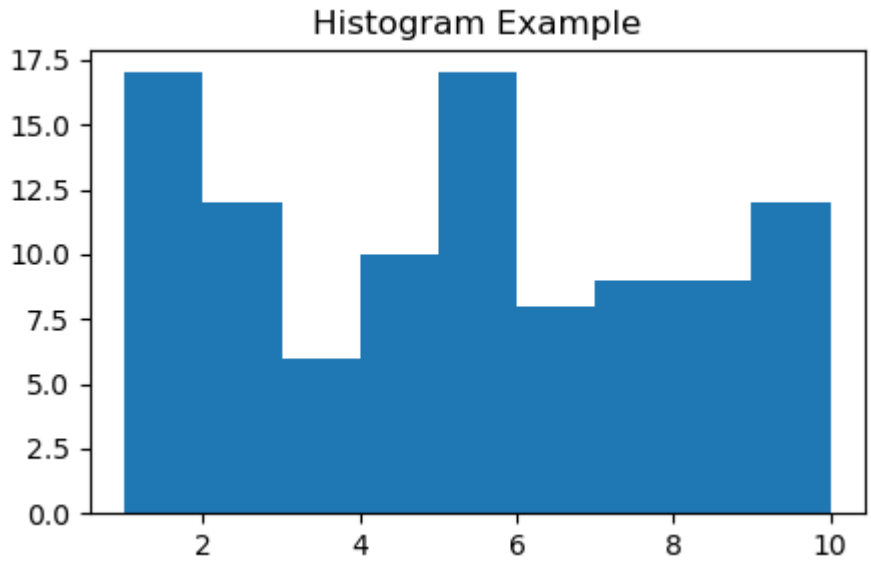
6. Histogram

A histogram is a graph used to show the distribution of numerical (continuous) data by dividing it into intervals called bins.

- X-axis → Data intervals (bins)
- Y-axis → Frequency (count)
- Bars are connected (no gaps)
- Used to understand data distribution

In [122…

```
plt.figure(figsize=(5,3))  # figure size

#data
import random
data=[random.randint(1,10) for _ in range(100)]

plt.hist(data, bins=9)   # histogram chart
plt.title('Histogram Example')
plt.show()
```



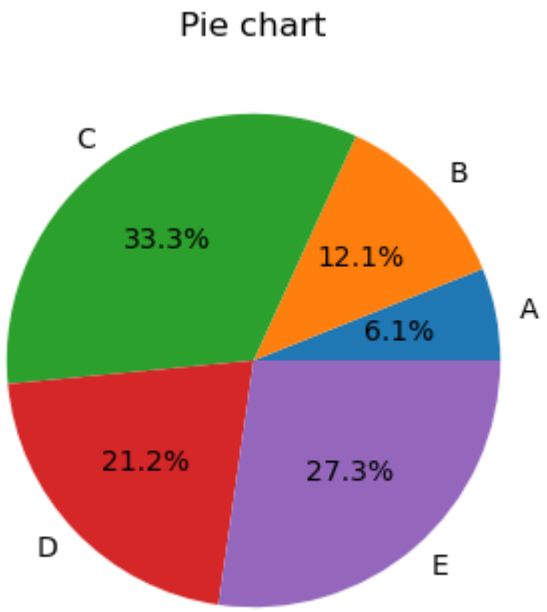
7. Pie Chart

A pie chart is used to show proportions or percentages of a whole using circular slices.

- Each slice represents a category
- Size of slice = proportion/value
- Total = 100%
- Best for percentage distribution

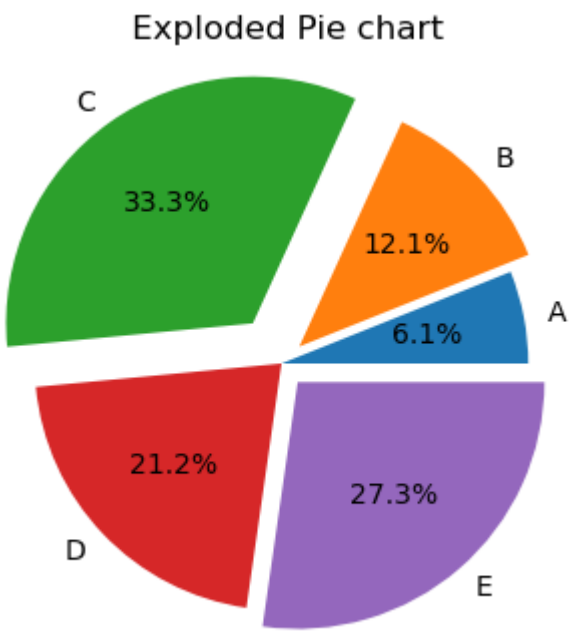
In [123…

```
plt.figure(figsize=(4,4))  # figure size

# data
categories= ['A','B','C','D','E']
sales= [10,20,55,35,45]

plt.pie(sales, labels=categories, autopct= '%1.1f%%')
plt.title('Pie chart')
plt.show()
```



In [124…

```
plt.figure(figsize=(4,4))  # figure size

# data
categories= ['A','B','C','D','E']
sales= [10,20,55,35,45]

plt.pie(sales, labels=categories, autopct= '%1.1f%%', explode=[0,0.1,0.2,0,0.1])

plt.title(' Exploded Pie chart')
plt.show()
```



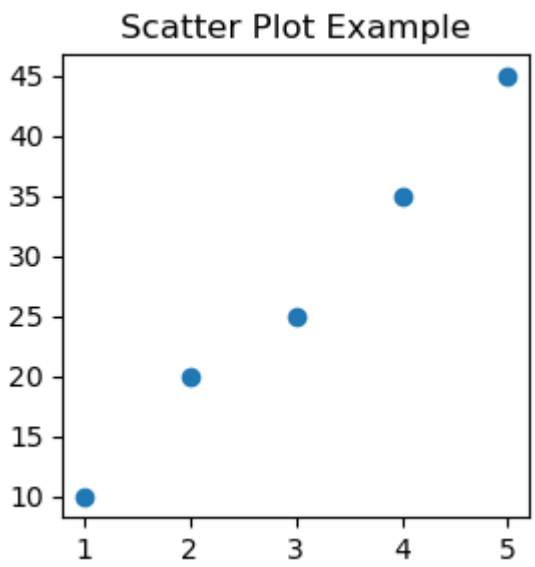
8. Scatter plot

Used to find relationship between variables

- Shows relationship between two variables (X & Y)
- Each point = one observation
- All points have same size
- Used to find correlation/trend

In [110…

```
plt.figure(figsize=(3,3))  # figure size

# data
x= [1,2,3,4,5]
y= [10,20,25,35,45]

plt.scatter(x,y)
plt.title('Scatter Plot Example')
plt.show()
```



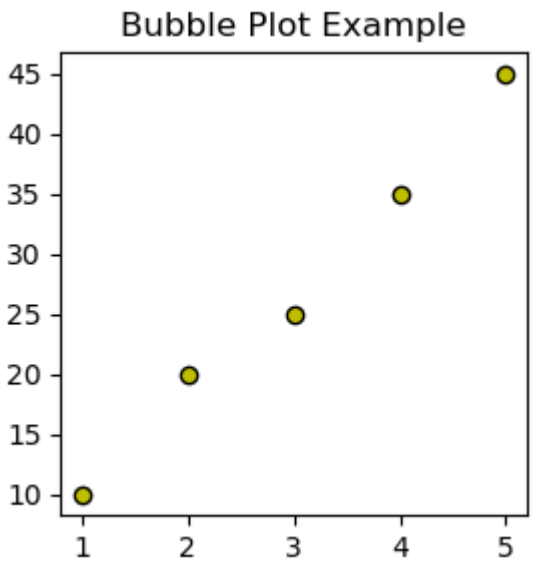
9. Bubble Plot

A bubble plot is a scatter plot where the size of each point represents a third variable.

- Extension of scatter plot
- Uses three variables (X, Y, Size)
- Bubble size represents extra data
- Used for multi-dimensional analysis

In [188…

```
plt.figure(figsize=(3,3))  # figure size

# data
x= [1,2,3,4,5]
y= [10,20,25,35,45]

plt.scatter(x,y, color= 'y', edgecolors='k') # edgecolors for outerline
plt.title('Bubble Plot Example')
plt.show()
```



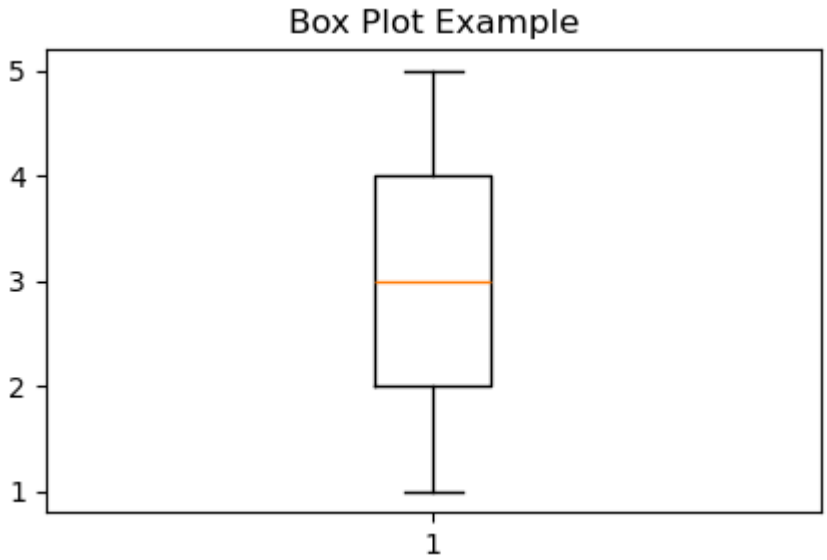
10. Box Plot

A box plot (also called a box-and-whisker plot) is a graphical way to summarize the distribution of data using five key values.

- Center (median)
- Spread (range & IQR)
- Skewness
- Outliers

In [92]:

```
plt.figure(figsize=(5,3))  # figure size

# data
x= [1,2,3,4,5]

plt.boxplot(x)
plt.title('Box Plot Example')
plt.show()
```



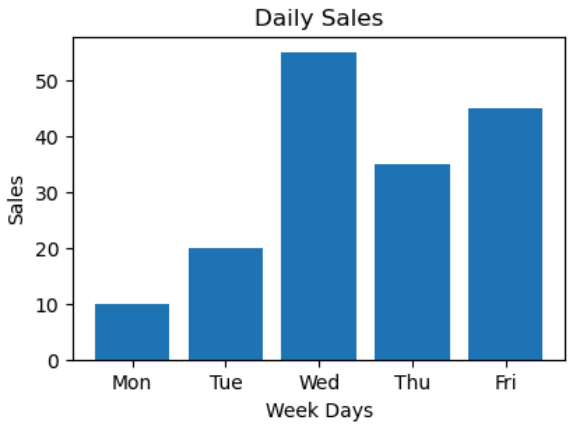
11. Subplot

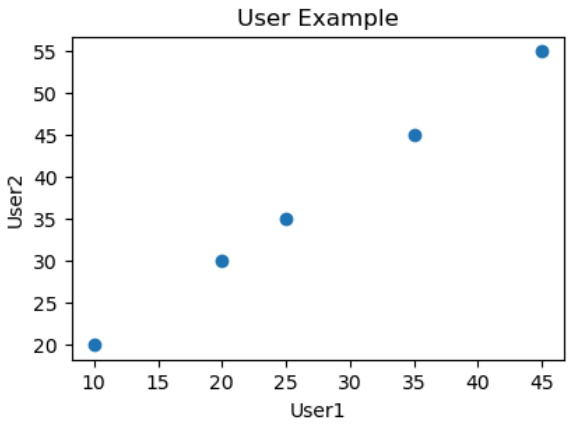
A subplot is used to display multiple plots in a single figure.

- Divides figure into a grid (rows × columns)
- Each plot is placed in a specific position
- Helps compare multiple graphs together

In [125…

```
# data-1 - bar chart
categories = ['Mon','Tue','Wed','Thu','Fri']
sales = [10, 20, 55, 35, 45]

# data-2 - scatter plot
y1 = [10, 20, 25, 35, 45]
y2 = [20, 30, 35, 45, 55]

plt.figure(figsize=(10,3))  # figure size

# first plot- bar chart
plt.subplot(1,2,1)   # row, column, position
plt.bar(categories, sales)
plt.title("Daily Sales")
plt.xlabel("Week Days")
plt.ylabel("Sales")

# second plot- scatter chart
plt.subplot(1,2,2)   # row, column, position
plt.scatter(y1, y2)
plt.title("User Example")
plt.xlabel("User1")
plt.ylabel("User2")

plt.show()
```





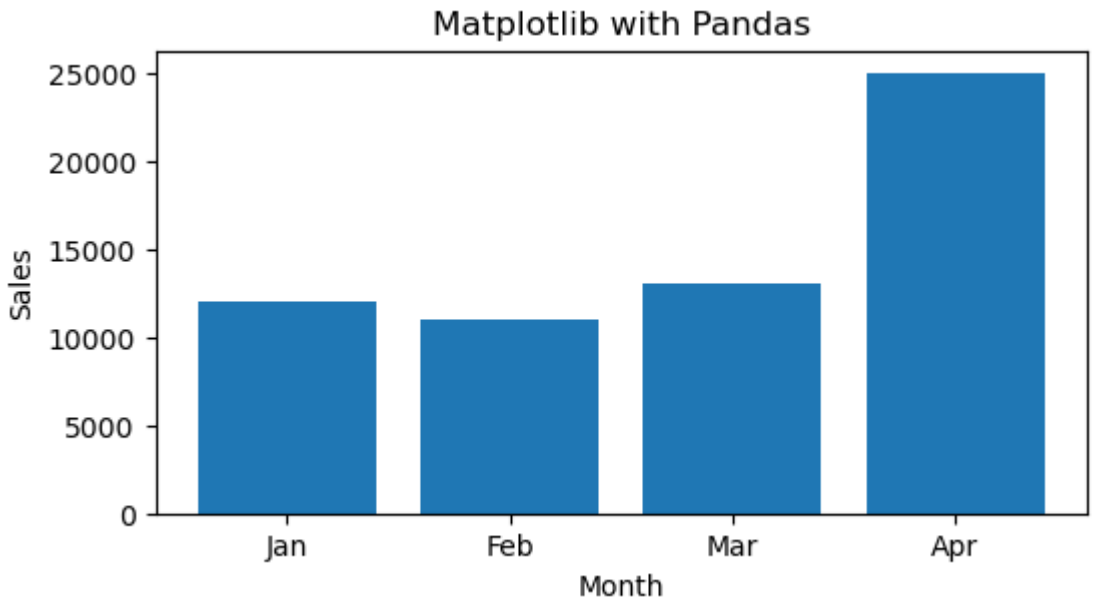
12. Matplotlib with Pandas - real data

In [42]:
```
# create df
import pandas as pd

# data
data = {
    'Month' : ['Jan', 'Feb', 'Mar', 'Apr'],
    'Sales' : [12000, 11000, 13000, 25000]
}

df = pd.DataFrame(data)
df
```

Out[42]:

	Month	Sales
0	Jan	12000
1	Feb	11000
2	Mar	13000
3	Apr	25000

In [127…
```
plt.figure(figsize=(6,3))  # fig. size

plt.bar(df['Month'],df['Sales'])
plt.title('Matplotlib with Pandas')
plt.xlabel('Month')
plt.ylabel('Sales')

plt.show()
```



13. Object-Oriented API

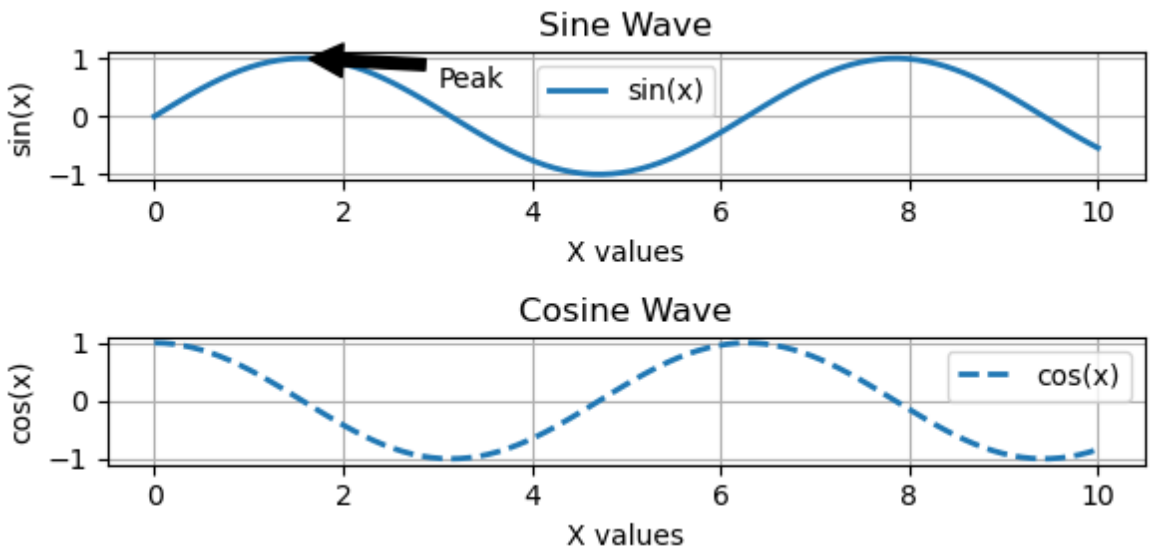
he Object-Oriented API in Matplotlib is a way of plotting where you control plots using objects like Figure and Axes instead of pyplot functions.

- Figure (fig) → Whole canvas (entire plot area)
- Axes (ax) → Individual plot (graph inside figure)

In [149…

```
# Generate data
x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = np.cos(x)

# Create figure and axes (2 subplots)
fig, axs = plt.subplots(2, 1, figsize=(6,3))

# ---- First Plot (Sine) ----
axs[0].plot(x, y1, label='sin(x)', linewidth=2)
axs[0].set_title("Sine Wave")
axs[0].set_xlabel("X values")
axs[0].set_ylabel("sin(x)")
axs[0].legend()
axs[0].grid(True)

# Add annotation (arrow sign)
axs[0].annotate('Peak', xy=(1.57, 1), xytext=(3, 0.5),
                arrowprops=dict(facecolor='black', shrink=0.05))

# ---- Second Plot (Cosine) ----
axs[1].plot(x, y2, label='cos(x)', linestyle='--', linewidth=2)
axs[1].set_title("Cosine Wave")
axs[1].set_xlabel("X values")
axs[1].set_ylabel("cos(x)")
axs[1].legend()
axs[1].grid(True)

# Adjust layout
```

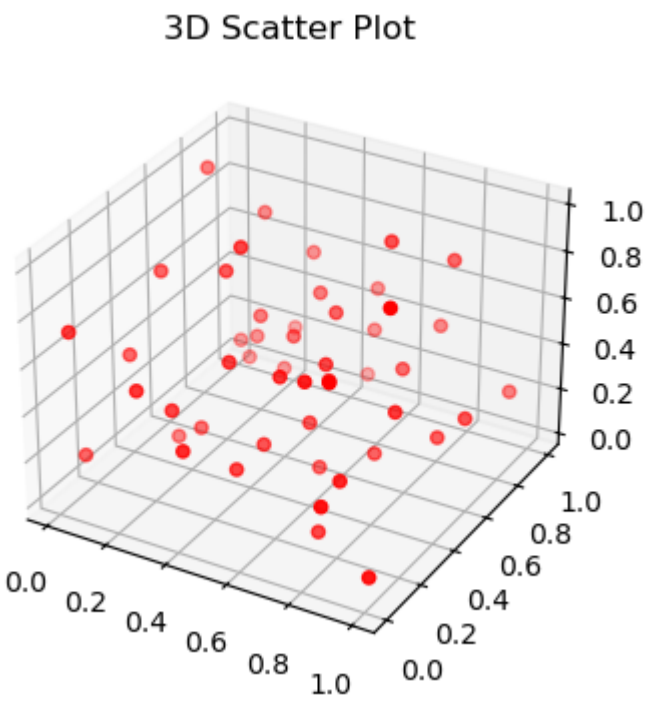
```
plt.tight_layout()

# Show plot
plt.show()
```



14. 3D Plot

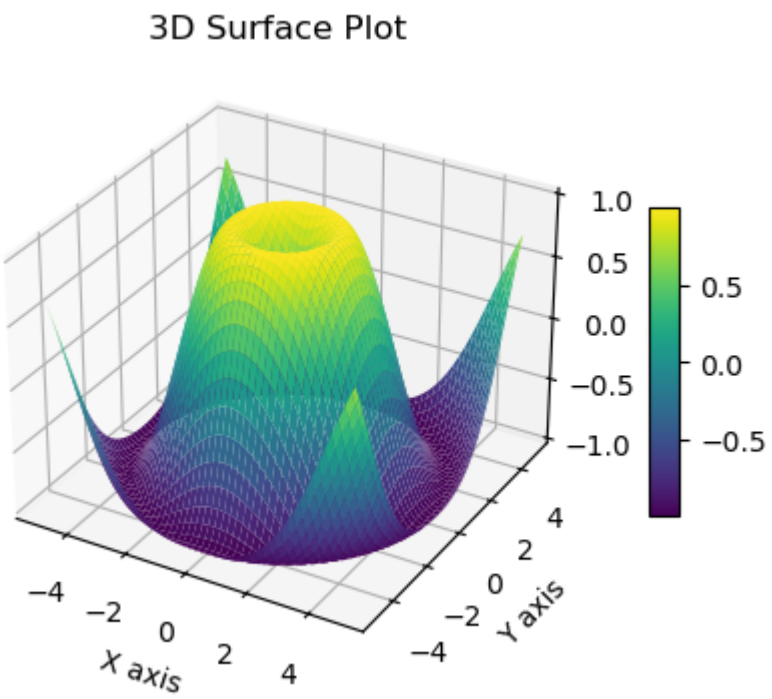
In [179…

```
fig = plt.figure(figsize=(4, 4))  # set figure size
ax = fig.add_subplot(111, projection='3d')  # 3D axes

x = np.random.rand(50)  # random x values
y = np.random.rand(50)  # random y values
z = np.random.rand(50)  # random z values

ax.scatter(x, y, z, c='red', marker='o')  # 3D scatter plot

ax.set_title("3D Scatter Plot")  # title

plt.show()  # display plot
```

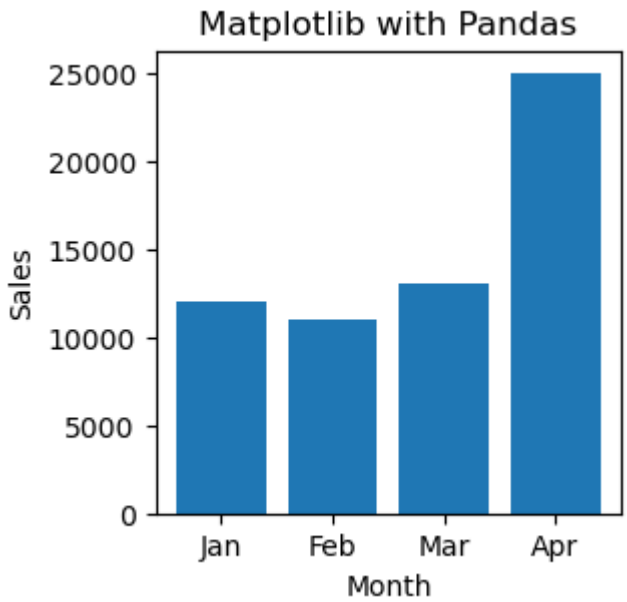


In [180…

```
# Import 3D toolkit
from mpl_toolkits.mplot3d import Axes3D

# Create data
x = np.linspace(-5, 5, 100)
y = np.linspace(-5, 5, 100)

X, Y = np.meshgrid(x, y)
Z = np.sin(np.sqrt(X**2 + Y**2))

# Create figure and 3D axes
fig = plt.figure(figsize=(8, 4))
ax = fig.add_subplot(111, projection='3d')

# Plot surface
surf = ax.plot_surface(X, Y, Z, cmap='viridis')

# Labels and title
ax.set_title("3D Surface Plot")
ax.set_xlabel("X axis")
ax.set_ylabel("Y axis")
ax.set_zlabel("Z axis")

# Add color bar
fig.colorbar(surf, ax=ax, shrink=0.5, aspect=10)

# Show plot
plt.show()
```



15. Save Charts

In [182…

```
plt.figure(figsize=(3,3))  # fig. size

# NOTE - don't use fig. size for better persentation or to save in good quality

plt.bar(df['Month'],df['Sales'])
plt.title('Matplotlib with Pandas')
plt.xlabel('Month')
plt.ylabel('Sales')

plt.savefig('Monthly_Sales_from_df.png')  # To Save Figure / Chart
# plt.savefig("graph.pdf")
# plt.savefig("graph.png")

plt.show()
```